Yanjie Yang

Ph.D. Candidate, Texas A&M University
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Research Fields

Labor · Public · Experimental

Interests: household liquidity, preference heterogeneity, maternal labor supply, and behavioral economics

Methods: structural labor-supply and consumption-saving models, discrete choice (DCM/DDCM), and counterfactual analysis

Education

Texas A&M University 2021–present

Ph.D. Candidate in Mathematical Economics; expected May 2027

Texas A&M University 2019–2021

M.S. in Economics

University of International Business and Economics, China 2015–2019

B.S. in Financial Engineering

Selected Quantitative Research

Graduate Researcher, Texas A&M University Aug 2021–Present

- Applied logistic regression to predict labor-market participation on 3M+ panel records; integrated web-scraped unemployment-insurance policy data.
- Implemented random forest and KNN for large-scale data imputation; used a Linux high-performance computing cluster to reduce computation time by 90%.

Job Market Paper

When Child Benefits Arrive: Payment Timing, Liquidity, and Mothers' Welfare

With Li Gan

October 2026 draft · Working paper updated November 5, 2026

The expanded 2021 Child Tax Credit raised benefits and paid half of them in monthly advances. Holding each family's annual credit fixed, we ask what the earlier payments were worth to married and single mothers in the Survey of Income and Program Participation. Relative to comparable women without children, mothers show no detectable change in employment and no detectable relative increase in bank balances; card debt fell, mostly among mothers who held some cash and carried debt, while low cash persisted among single mothers. A monthly consumption-saving model with income and childcare-cost risk, in which some mothers save and others spend all their cash, is estimated by maximum likelihood on each mother's records with patience set at a standard value. The model-based spender probability averages 31 percent for single mothers and 2 percent for married mothers. Relative to paying the same credit at filing, the 2021 calendar was worth $191 per married mother and $987 per single mother, and $993 and $1,820 for mothers with $1,000 or less in liquid assets. For the average married and single mother, each added credit dollar paid only at filing was worth 92 and 60 cents, against 102 and 96 cents paid as in 2021. The single-mother value is conditional on the spender description: valuing every mother as a saver gives $476, and impatient savers with conventional discount factors, valued at patient rates, give $483 to $539, about half as large.

Working Papers

An LLM-RAG Approach for Healthy Eating Index-Informed Personalized Food Recommendations

With Grace Melo, Rudy Nayga, Azlan Zahid, and Yibin Wang

Under review

Develops a Healthy Eating Index-informed retrieval-augmented generation framework linking large language models to NHANES and the Food Patterns Equivalents Database. Evaluates simulated food substitutions and additions using standardized nutrition data and estimated HEI contributions.

AI-Assisted Personalized Food Recommendations and Diet Quality: Evidence from a Randomized Controlled Trial

With Grace Melo, Rudy Nayga, Azlan Zahid, and Yibin Wang

In a pre-registered randomized trial (AEARCTR-0015970), 839 U.S. adults received either a mobile app delivering AI-personalized meal recommendations or an otherwise identical app delivering general healthy-eating guidance. After seven weeks we find no effect on total diet quality: +0.74 HEI-2020 points (95% CI −3.49 to +4.98). Whole grains, one of thirteen components, is 1.32 points higher under treatment ($q = 0.080$). Willingness to pay rises \$1.29 per standard deviation of perceived relevance ($p = 0.001$) but is flat in measured dietary change (−\$0.15, $p = 0.49$). Consumers pay for how personalized an app feels, not for what it measurably does.

Research in Progress

Childcare Support and Maternal Employment: Income and Liquidity Channels in the UK's 2023 Reform

With Li Gan

A Life Cycle Model on Mother's Labor Participation and Mental Health

With Di Fang and Yong Liu

Studies maternal employment and mental health using PSID panel data, a dynamic discrete-choice model (**DDCM**) of job search, and counterfactual policy simulations.

Monthly or Annual? Measured Liquidity and the Welfare Economics of Child Tax Credit Disbursement Frequency

With Li Gan

Conference Presentations and Selected Papers

Advances with Field Experiments Conference, Chicago, IL Sep 2026

AAEA Annual Meeting, Kansas City, MO Jul 2026

AAEA Annual Meeting, Denver, CO Jul 2025

Symposium for Agricultural and Applied Economics Research, College Station, TX May 2025

American Public Health Association Annual Meeting, Atlanta, GA Nov 2023

American Academy for Health Behavior Annual Meeting, San Francisco, CA Mar 2023

American Public Health Association Annual Meeting, Boston, MA Nov 2022

Selected paper

ASHEcon Annual Meeting, Minneapolis, MN Jun 2026

Teaching Experience

Texas A&M University

Co-Instructor Spring 2023

AGEC 317: Economic Analysis for Agribusiness Management

Taught econometric methods, led hands-on Excel labs, and mentored student projects.

Received positive student feedback that AGEC 317 supported summer internship work and informed career decisions.

Graduate Teaching Assistant

AGEC 447: Food and Agricultural Price Analysis Fall 2023, 2024

Senarath Dharmasena

AGEC 453: International Agribusiness Marketing Spring 2024

Emiliano Lopez Barrera

AGEC 448: Agricultural Commodity Futures Spring 2022, 2024

Dimitry Vedenov

AGEC 344: Food and Agricultural Law Spring 2022; Fall 2023

Fred Boadu

AGEC 317: Economic Analysis for Agribusiness Management Fall 2021, 2022

Kuo Liu and Dominic Vieira

AGEC 330: Financial Management in Agriculture Summer 2022

Chrystol Thomas

AGEC 413: Agricultural Cooperatives Fall 2021

John Park

Fellowships and Travel Grants

HEEP Fellowship, Texas A&M University 2021

Departmental AAEA Travel Grants 2022, 2025, 2026

GSA Travel Grants 2022, 2026

Academic Service

Student Lead, AGEC Seminar Committee Aug 2023–Oct 2024

Texas A&M University

Coordinated weekly academic seminars with 20+ invited speakers from across the U.S.; managed logistics, communications, speaker invitations, and a $10,000 budget for event operations and speaker accommodations.

Travel Grant Officer, AGEC Graduate Student Association Sep 2022–Aug 2023

Texas A&M University

Shared application information and administered graduate-student travel grant budgets.

Social Media Volunteer Sep 2022–Dec 2023

AAEA Committee on Women in Agricultural Economics

Created content highlighting women scholars' achievements in agricultural economics.

Professional and Entrepreneurial Experience

Business Analyst Intern, Tesla Academy, Tesla Dec 2018–Apr 2019

Beijing, China

Automated sales-staff assignment reporting with Excel VBA, reducing analysis time from 30 minutes to 1 minute. Created and shared Model 3 training materials on MS Yammer to support sales-staff onboarding.

Cofounder, Chuangzhi Consulting Studio Jan 2016–Dec 2018

University of International Business and Economics, Beijing, China

Helped a startup team secure US$1.5 million in funding from Baidu Inc. by developing its business plan. Advised student startup teams on business planning, funding, and marketing; four teams earned first- or second-place awards in the National Innovation Competition. Registered the studio as a company and participated in the National Innovation Training Program.

Methods and Technical Skills

Causal inference: difference-in-differences, regression discontinuity, instrumental variables, bootstrap inference

Structural methods: consumption-saving, labor-supply, and life-cycle models; **DCM** and **DDCM**; simulated method of moments, maximum likelihood, dynamic programming

Statistical machine learning: logistic regression, Random Forest, Gradient Boosting, nearest neighbors, PCA, LDA, SVM, cross-validation

Software: Python, R, SQL, scikit-learn, H2O, MATLAB, Excel VBA, Linux, LaTeX, EViews, Gretl

Experiments and computing: experimental design, RCTs, A/B testing, HPC/parallel computing

Languages: Mandarin Chinese (native), English (fluent)

Additional Training

Decision Making and Reinforcement Learning Aug 2025

Columbia University/Coursera

Machine Learning 2022

Stanford/Coursera certificate

Learning SSH; Learning Linux Command Line 2022

LinkedIn Learning

English Language Proficiency Exam certificate 2021

Texas A&M University

References

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Rudy Nayga	rudy.nayga@ag.tamu.edu